Acyclicity and simultaneous protection of individual core maxima

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September 20, 2026

Abstract

We characterize the sector interaction graphs that support a uniform fraction of every player's maximum attainable core payoff. Fix a connected graph on $K \geq 2$ equal-sized sectors, with nonnegative pair values on its edges; only full K-sector teams produce value, and coalition worth is integral matching worth. Conditional on a nonempty core, both the full leximin guarantee and the best simultaneous core guarantee have sharp size-independent constant $1/K$ for trees and zero for graphs containing a cycle. The tree result follows from an entire-core cut representation and a clipping argument valid at every lexicographic level. For a triangular family with pair entries in $\{0, 1, 2\}$, we characterize the entire core as a simplex and prove that the best simultaneous fraction is exactly $1/(2q + 3)$. Selective subdivision preserves a unit partial-matching shortfall and transfers the impossibility to every cyclic allowed graph, while retaining a strictly positive core allocation. Thus the cyclic failure persists even when the allocation rule is chosen optimally. Complete proofs and reproducible rational and exact-solver checks are supplied. Historical priority of the precise classification remains qualified.

Keywords: assignment game; core; leximin; individual ideal; simultaneous protection; sector graph.

1 Introduction

A core allocation protects coalitions against profitable deviation. It need not protect each individual relative to what that person could receive at another stable allocation. For a core $\mathcal{C}$ and individual ideal $H_i = \max_{y \in \mathcal{C}} y_i$, we study two questions: how much of H_i full leximin guarantees, and whether a different core allocation can protect all players better. The latter optimizes over the entire core, not over a specified selection rule.

The same sharp graph boundary governs both questions in the model studied here. Trees guarantee $1/K$; every cyclic allowed graph admits growing markets with vanishing best simultaneous protection, even though the core contains a strictly positive point. The poor recipient in the simultaneous statement may depend on the chosen allocation. We do not claim that a single player receives little at every core point: each ideal is attained.

This manuscript consolidates the earlier chain–cycle separation with its subsequent tree theorem, general cyclic transfer, and simultaneous-protection strengthening. The triangle is the same construction throughout. The simplex representation and the all-selection conclusion strengthen its interpretation; they are not an additional independently discovered family. The results concern worst-case individual guarantees, not aggregate welfare loss or typical markets.

Related work. Multisided assignment cores have a substantial literature, including Quint [1] and Lucas [2]. The strict supplier–firm–buyer model is associated with Stuart [3]; Atay and Solymosi [4],

Definition 3 and equations (1)–(2), give a precise later formulation allowing signed adjacent-pair values and replacing negative triple totals by zero. Their core and bargaining-set results concern stability sets, rather than the coordinate-ideal fraction studied here. Nonnegative chain values are a subclass of that model. Generalized assignment markets in which smaller teams also produce value have additional coalition constraints and must be distinguished from our strict model [5].

Ideal-based core protection is explicit in Núñez and Rafels [6], Definition 6, in their bilateral bargaining analysis. A unit-capacity path in Gangam et al. [7], Section 3.1.4, has uniform equitable payoff $1/r$ on its r edges. Its full core is the unit simplex, so each ideal is one: vanishing protection in a broader game class follows immediately. That path uses a productive coalition whose size grows with r. Our construction instead uses only strict triples and pair values in $\{0, 1, 2\}$.

Matching duality, tight-team cut coordinates, and closure of difference constraints under minimum are established tools. The tree proof below gives a direct clipping argument; no new general optimization method is claimed. Núñez and Rafels' bilateral half-optimal core is also a direct precedent for simultaneous ideal protection, rather than just for comparing named selectors [6]. Tree decomposition and tree/cycle balancedness results in models with productive partial-sector coalitions [5] must not be identified with our classification conditional on a nonempty strict full-team core.

Extreme-core descriptions for assortative multisided markets [9] and supermodular markets with local monotonicity [10] impose restrictions different from nonnegative pair additivity. We make no assertion that arbitrary extreme-core or transportation representations preserve the entire labeled payoff core, the raw sorted selector, and our pair-value restrictions. The distinct 2023 manuscript on extreme core points [11] is now available for comparison. Its Theorem 2 redescribes the full core through cumulative sums of auxiliary-game core allocations, retaining nonnegativity and the triple inequalities outside a central strip. Theorems 3–4 simplify this under additional restrictions. These are core descriptions; they do not state the vanishing family or its full-leximin ideal ratio. The original texts of [1, 2, 3] remain access gaps at the date of this version. No theorem below relies on an uninspected statement from these originals.

2 Model and main classification

Fix a connected simple graph $G = (V, E)$ with $K = |V| \geq 2$. Each sector $v \in V$ contains players (v, j), $1 \leq j \leq m$, for a common finite $m \geq 1$. An allowed edge uv carries a nonnegative matrix A_{uv}; zero matrices are permitted. All full teams $t = (j_v)_{v \in V}$ are available, with worth

$$w(t) = \sum_{uv \in E} A_{uv}(j_u, j_v).$$

Only full teams produce value. For any coalition S, its worth $v(S)$ is the maximum sum of worths of player-disjoint full teams contained in S. In particular every singleton has worth zero. Set $T = v(N) > 0$ and require that the entire core $\mathcal{C}$ be nonempty. Equivalently,

$$\mathcal{C} = \{p \geq 0 : \sum_i p_i = T,\ \sum_{i \in t} p_i \geq w(t) \text{ for every full team } t\}. \tag{1}$$

Indeed these conditions imply every coalition inequality by summing over its integral matching and retaining nonnegative unmatched payoffs.

The compact convex set $\mathcal{C}$ has coordinate maxima H_i. Let x be its full ascending-sorted leximin allocation; all labeled coordinates, including permanent zeros, remain in the optimization. Existence follows by successive maximization of order statistics on compact optimal faces. Uniqueness follows

by averaging two distinct purported optima: at their first differing payoff level the average reduces its multiplicity without changing lower levels. Define

$$c(G) = \inf_M \min_{i:H_i>0} \frac{x_i}{H_i}, \qquad \rho(M) = \max_{p\in\mathcal{C}} \min_{i:H_i>0} \frac{p_i}{H_i}, \qquad R(G) = \inf_M \rho(M),$$

where M ranges over the stated markets on fixed G, allowing m to grow. Thus $c(G) \le R(G)$. The ratios are formed only when $H_i > 0$; other coordinates are not deleted from the core or leximin problem.

Theorem 2.1. *For every fixed connected allowed graph on $K \ge 2$ sectors,*

$$c(G) = R(G) = \begin{cases} 1/K, & G \text{ is a tree,} \\ 0, & G \text{ contains a cycle.} \end{cases}$$

For every cyclic G and integer $q \ge 1$, the upper witnesses have $3q+1$ players per sector, pair entries in $\{0,1,2\}$, and a strictly positive core allocation. One target has ideal 1 and full leximin payoff $1/[K(q+1)]$, while $\rho(M) \le 1/(2q+2)$. For the source triangle, the exact stronger formula is $\rho(M_q) = 1/(2q+3)$.

Every fixed finite market still has $\rho(M) > 0$: average one core maximizer for each of its n_+ positive ideals to obtain $p_i \ge H_i/n_+$. The zero in the theorem is an infimum over sizes, not a finite-market zero. Also, the theorem concerns allowed edges, not strictly positive matrices on every allowed edge. A single-sector edge-sum model has no positive-worth instance and is excluded.

3 Trees: entire-core coordinates and clipping

Root the tree at an arbitrary sector r. Choose an optimal perfect matching on each edge and propagate labels from the root so that each is diagonal. There is no conflicting propagation on a tree. Write $d_{ab}(j) = A_{ab}(j,j)$ for an oriented parent–child edge $a \to b$. Any full-team matching projects to edge matchings; conversely the diagonal full-team matching attains the sum of these edge optima. Hence $T = \sum_j \sum_{ab\in E} d_{ab}(j)$. Nonnegative weights and equal sizes permit completion of partial matchings without loss. Summing optimal nonnegative bilateral dual covers over incident edges gives a full-team cover of total T, so tree cores are nonempty. Every diagonal team is tight throughout the core.

For a nonroot vertex b, let S_b be its descendant subtree, $s_b = |S_b|$, and define

$$U_b^p(j) = \sum_{v\in S_b} p_v(j) - \sum_{e \text{ internal to } S_b} d_e(j), \qquad U_r^p(j) = 0.$$

Lemma 3.1. *The entire core is in bijection with the cut vectors satisfying*

$$U_b(i) - U_b(j) \ge A_{ab}(j,i) - d_{ab}(j), \tag{2}$$

$$p_v(j) = U_v(j) - \sum_{b \text{ child of } v} U_b(j) + \sum_{b \text{ child of } v} d_{vb}(j) \ge 0. \tag{3}$$

Proof. For necessity, take label i on S_b and label j on its complement in a full team, and subtract the tight diagonal team at j. The single cut edge is ab; internal diagonal terms cancel to give (2). Subtracting child-subtree sums gives (3); at the root use diagonal tightness. Conversely reconstruction telescopes at each fixed label to the diagonal team value and therefore gives total T. At arbitrary labels,

$$\sum_v p_v(i_v) = \sum_{a\to b} [U_b(i_b) - U_b(i_a) + d_{ab}(i_a)] \ge \sum_{a\to b} A_{ab}(i_a, i_b).$$

Thus every full team is covered. Nonnegativity and (1) imply core membership. Descendant sums recover the cut coordinates, proving bijectivity. □

Lemma 3.2 (Clipping)**.** *For $x, y \in \mathcal{C}$ and $\beta \geq 0$, set $Z_b(j) = \min\{U_b^x(j), U_b^y(j) + s_b\beta\}$ for $b \neq r$, and $Z_r = 0$. The reconstructed z belongs to $\mathcal{C}$ and satisfies*

$$z_v(j) \geq \min\{x_v(j), y_v(j) + \beta\} \quad (v \neq r), \qquad z_r(j) \geq \max\{x_r(j), y_r(j) - (K-1)\beta\}.$$

Proof. Difference inequalities $u(i) - u(j) \geq c_{ij}$ are closed under uniform translation and coordinate-wise minimum: both candidate values at i are at least their respective value at j plus c_{ij}. Thus Z satisfies all structural inequalities. At a nonroot vertex, if its selected cut is the x cut, each child cut is at most its x cut and reconstruction gives $z_v \geq x_v$. If it is the shifted y cut, use $s_v = 1 + \sum_b s_b$ to obtain $z_v \geq y_v + \beta$. These cases include ties. At the root, upper-bound each child cut first by its x value and then by its shifted y value; the child subtrees have total size $K - 1$. This gives both root inequalities. All reconstructed payoffs are nonnegative; Lemma 3.1 finishes the proof. □

Fix a target at the root sector and put $\alpha = x_i$ for full leximin x. If $H_i > K\alpha$, choose an ideal-attaining y and $\alpha < \beta < H_i/K$. Clipping gives $z_i \geq H_i - (K-1)\beta > \beta > \alpha$. Every old coordinate at most α weakly increases, and every old coordinate above α remains strictly above it. Thus, for each $\gamma \leq \alpha$, the number of coordinates below γ cannot increase, and the set of coordinates at most α strictly shrinks because the target leaves it. Decreases above α do not affect this comparison. If a lower level increases, its first changed multiplicity decreases; otherwise the multiplicity of α decreases. Either case strictly improves the sorted vector, a contradiction. Consequently $x_i \geq H_i/K$, even when $x_i = 0$. Re-rooting the tree and relabeling its optimal matching change only the coordinates used in the proof, not the chosen leximin allocation x. The target was arbitrary, so this same x protects every player and $\rho(M) \geq 1/K$ as well.

For sharpness take $m = 1$ and edge values totaling one. The sole full team has worth one and every proper coalition has worth zero. Its core is the K-coordinate unit simplex; $H_i = 1$, leximin is uniform, and $\rho = 1/K$. This proves both tree equalities. It also gives the sharp distance bound $\|x - y\|_1/T \leq 2(1 - 1/K)$ for any tree-core point y, since $\min(x_i, y_i) \geq y_i/K$ and both totals are T.

4 A cyclic family with vanishing protection

Fix an integer $q \geq 1$. Use the common team-label set

$$\mathcal{T} = \{L_0\} \cup \{O_s, L_s, R_s : 1 \leq s \leq q\}.$$

For every label t, create players a_t, b_t, c_t in the respective sectors. Each sector has $3q + 1$ players. Define $w(a_t, b_u, c_v) = A_{tu} + B_{tv} + C_{uv}$ by the following nonzero entries; all unspecified pair entries are zero.

Pair entries	Range	Value
C_{L_s,L_s}	$0 \leq s \leq q$	1
C_{R_s,R_s}	$1 \leq s \leq q$	1
A_{O_s,O_s}, B_{O_s,O_s}	$1 \leq s \leq q$	2
A_{O_s,R_s}, B_{O_s,L_s}	$1 \leq s \leq q$	2
C_{R_s,L_s}, C_{L_{s-1},O_s}	$1 \leq s \leq q$	1

There are $8q + 1$ nonzero entries. The diagonal perfect matching consists of (a_t, b_t, c_t) for all $t \in \mathcal{T}$. Each L or R triple has value one and each O triple has value four, so this matching is worth $6q + 1$.

4.1 Two explicit full-core covers

Put $\tau = 1/[3(q+1)]$. Define allocations u and y in (a, b, c) order:

team	u	y
L_s	$(\tau,\ 1-(3s+2)\tau,\ (3s+1)\tau)$	$(0,0,1)$
R_s	$(\tau,\ 1-2\tau,\ \tau)$	$(0,1,0)$
O_s	$(4-(3s-1)\tau,\ \tau,\ (3s-2)\tau)$	$(3,0,1)$.

(4)

The L row uses $0 \le s \le q$; the other rows use $1 \le s \le q$. Every coordinate of u is at least $\tau > 0$, and y is nonnegative. The smallest L-sector b coordinate is attained at $s = q$ and equals τ; the smallest O-sector c coordinate is attained at $s = 1$ and also equals τ. Each diagonal triple receives its value at both points, so both allocations have total $6q + 1$.

We check every triple cover, including all mixed-team triples. If its a player has an L or R label, its A and B entries are zero. The possible positive C pairs and the associated two-player sums are

(b, c) labels	$u_b + u_c$	$y_b + y_c$
(L_s, L_s)	$1-\tau$	1
(R_s, R_s)	$1-\tau$	1
(R_s, L_s)	$1+(3s-1)\tau$	2
(L_{s-1}, O_s)	$1-\tau$	1

All four pair values are one. Adding $u_a = \tau$ or $y_a = 0$ covers the triple. Zero C entries are covered by nonnegativity.

Now suppose its a player is a_{O_s}. If at most one of its A and B entries is positive, the triple value is at most three, covered by $u_{a_{O_s}} \ge 3 + 4\tau$ and $y_{a_{O_s}} = 3$. If both entries are positive, exactly four cases remain:

(b, c) labels	w	$u_a + u_b + u_c$	$y_a + y_b + y_c$
(O_s, O_s)	4	4	4
(O_s, L_s)	4	$4+3\tau$	4
(R_s, O_s)	4	$5-3\tau$	5
(R_s, L_s)	5	5	5

Since $\tau \le 1/6$, all are covers. The cases also show that every triple value is at most five.

The matching gives worth at least $6q + 1$, while either nonnegative cover bounds every matching by its total $6q + 1$. Hence the true integral worth is $T = 6q + 1$ and both u, y belong to the entire core. No fractional matching value is substituted for coalition worth.

4.2 A partial matching determines the target

For each $s = 1, \dots, q$, take the triples

$$N_s = (a_{O_s}, b_{R_s}, c_{L_s}), \qquad E_s = (a_{L_{s-1}}, b_{L_{s-1}}, c_{O_s}).$$

Their values are five and one, respectively. They form a player-disjoint partial matching: the used a labels are the O labels and $L_0, \dots, L_{q-1}$; the used b labels are the R labels and the same L labels; the used c labels are $L_1, \dots, L_q$ and the O labels. Its total is $6q$. Its unmatched players are exactly

$$U = \{c_{L_0}, a_{L_q}, b_{L_q}\} \cup \{a_{R_s}, b_{O_s}, c_{R_s} : 1 \le s \le q\}, \qquad |U| = 3q + 3. \tag{5}$$

Summing the partial-matching inequalities and subtracting from efficiency gives, at every core point p,

$$\sum_{a \in U} p_a \leq 1. \tag{6}$$

Thus a common payoff floor cannot exceed $1/(3q+3) = \tau$. The point u attains this floor, so the first global leximin level is exactly τ. At every point of the first optimal face, all coordinates are at least τ, and $|U|\tau = 1$. Inequality (6) therefore fixes every coordinate of U at τ on that entire face. In particular, the full leximin target c_{L_0} is fixed at τ through every later refinement. We do not need, and do not assert, that u is the complete final leximin vector.

The diagonal L_0 triple is tight at every core point and has value one. Nonnegativity implies $H_{c_{L_0}} \leq 1$, while the cover y attains one at that coordinate. Consequently

$$H_{c_{L_0}} = 1, \qquad x_{c_{L_0}} = \frac{1}{3(q+1)}.$$

This proves the triangular nominal-leximin conclusion. The next section characterizes the best simultaneous protection over the entire core.

5 The entire triangular core and optimal simultaneous protection

5.1 A simplex representation

All weak a-players, meaning those at L or R labels, have identical productivity. Replace the weak a-player in a tight diagonal triple by any other weak a-player. The resulting inequality compares their payoffs; reversing the replacement proves equality. Denote their common payoff by $\alpha \geq 0$. Write

$$l_s = p_{c_{L_s}} \ (0 \leq s \leq q), \qquad r_s = p_{c_{R_s}},\ d_s = p_{b_{O_s}},\ o_s = p_{c_{O_s}} \ (1 \leq s \leq q).$$

Diagonal tightness gives

$$p_{b_{L_s}} = 1 - \alpha - l_s, \quad p_{b_{R_s}} = 1 - \alpha - r_s, \quad p_{a_{O_s}} = 4 - d_s - o_s.$$

The critical triples E_s, N_s respectively give $o_s \geq l_{s-1}$ and $l_s \geq \alpha + r_s + d_s + o_s$. Put $e_s = o_s - l_{s-1}$, $f_s = l_s - \alpha - r_s - d_s - o_s$, and $h = p_{b_{L_q}}$. These variables satisfy

$$(q+1)\alpha + l_0 + \sum_{s=1}^{q}(d_s + r_s + e_s + f_s) + h = 1, \qquad \alpha, l_0, d_s, r_s, e_s, f_s, h \geq 0. \tag{7}$$

Proposition 5.1. *Equation* (7)*, with recursive reconstruction*

$$o_s = l_{s-1} + e_s, \qquad l_s = l_{s-1} + \alpha + d_s + r_s + e_s + f_s,$$

characterizes the entire core of M_q. In particular the core is affinely isomorphic to a simplex of dimension $4q+2$.

Proof. Necessity follows from the displayed tight teams and inequalities. For sufficiency, l_s is nondecreasing and $l_q = 1 - \alpha - h \leq 1 - \alpha$, proving nonnegativity of all b_L payoffs. Also $r_s \leq l_s - \alpha \leq 1 - 2\alpha$, giving nonnegativity of b_R. Since $d_s + o_s \leq l_s - \alpha - r_s \leq 1$, every strong a-payoff is at least three. The other coordinates are visibly nonnegative and diagonal sums give efficiency.

For a weak a-player, the positive C edges of types $(L_s, L_s), (R_s, R_s), (R_s, L_s), (L_{s-1}, O_s)$ have total triple payoffs

$$1, \quad 1, \quad 1 + l_s - r_s, \quad 1 + o_s - l_{s-1},$$

respectively, all at least their worth one. If a strong a_{O_s} has at most one positive adjacent A or B entry, its own payoff of at least three covers the triple's worth of at most three. When both are positive, the four (b, c) label pairs $(O_s, O_s), (O_s, L_s), (R_s, O_s), (R_s, L_s)$ have worths $4, 4, 4, 5$ and respective payoff sums

$$4, \quad 4 + l_s - o_s, \quad 5 - \alpha - r_s - d_s, \quad 5 + l_s - \alpha - r_s - d_s - o_s.$$

They cover their worths because $l_s \geq o_s$, $\alpha + r_s + d_s \leq l_s \leq 1$, and $f_s \geq 0$. Zero-edge cases are covered by nonnegativity. This exhausts all full triples and proves core membership by (1).

The parameters are recovered from the payoffs and differences already specified, so reconstruction is injective. After replacing α by $(q+1)\alpha$, equation (7) has $4q+3$ nonnegative barycentric coordinates summing to one, hence dimension $4q + 2$. □

5.2 All ideals and the exact best fraction

The simplex representation yields

$$H_{a_{L_s}} = H_{a_{R_s}} = \frac{1}{q+1}, \qquad H_{a_{O_s}} = 4, \qquad H_{b_t} = H_{c_t} = 1 \quad \text{for every label } t. \tag{8}$$

For the weak a upper bounds use (7); for strong a use diagonal tightness. The reconstructed formulas bound each b, c coordinate by one: in particular $o_s, d_s, r_s \leq l_s \leq 1$. Attainment uses separate simplex vertices. Set only $\alpha = 1/(q+1)$ for weak a ideals; only $h = 1$ for all strong a and weak b ideals; only $d_s = 1$ for b_{O_s}; only $r_s = 1$ for c_{R_s}; and only $l_0 = 1$ for all c_L, c_O. No simultaneous attainment is assumed.

For any partial matching P in any admissible market, let U be its unmatched players and $\delta = T - w(P)$. Every core point satisfies $\sum_{i \in U} p_i \leq \delta$. Consequently

$$\rho(M) \leq \frac{\delta}{\sum_{i \in U} H_i} \qquad \text{if } \sum_{i \in U} H_i > 0. \tag{9}$$

This follows directly by summing the desired floors over U; even lower bounds on those ideals suffice for a weaker upper certificate.

For the triangular critical matching, $\delta = 1$ and the unmatched set in (5) has $q+1$ weak a-players, each of ideal $1/(q+1)$, and $2q+2$ players in sectors b, c, each of ideal one. Thus $\sum_U H_i = 2q + 3$ and $\rho(M_q) \leq 1/(2q+3)$. For equality put $t = 1/(2q+3)$ and choose

$$\alpha = \frac{t}{q+1}, \quad l_0 = h = t, \quad d_s = r_s = t, \quad e_s = f_s = 0.$$

Equation (7) holds. Every b, c payoff is at least t: $l_s, o_s \geq t$, $d_s = r_s = t$, $b_{L_s} \geq h = t$, and $b_{R_s} = 1 - \alpha - t \geq t$. Weak a-payoffs equal tH_i, and strong a-payoffs are at least $3 \geq 4t = tH_i$. Hence

$$\rho(M_q) = \frac{1}{2q+3}.$$

Already $q = 1$ gives a twelve-player market with best fraction $1/5 < 1/3$. The optimizing point is strictly positive. A lottery over core allocations cannot improve the simultaneous guarantee in expected payoffs: its mean remains in the convex core. This observation makes no claim about other criteria for evaluating lotteries.

6 Selective subdivision and every cyclic allowed graph

6.1 A transfer lemma with explicit hypotheses

Take a strict k-sector market with an optimal perfect matching relabeled diagonally, worth T. Suppose edge uv has matrix $B(i,j)$, with i at u. Choose $D_j \geq \max_i B(i,j)$. Insert a sector d, replacing uv by ud with matrix B and dv with diagonal entries D_j and off-diagonal zeros. For an old team t using u-label i and v-label j, and a new label h, its new worth is

$$w_1(t,h) = w_0(t) - B(i,j) + B(i,h) + D_h \mathbf{1}_{h=j}.$$

For every old core point z, the unshifted lift (z, D) covers every new team. If $h = j$ this follows from the old inequality with the bonus added. If $h \neq j$, use $D_h \geq B(i,h)$ and $B(i,j) \geq 0$. The extended diagonal matching and this cover have the same total $T' = T + \sum_j D_j$, proving integral optimality and lifted core membership. In particular all new diagonal teams are tight throughout the new core.

If z has floor τ, let $t = k\tau/(k+1)$. Subtract t/k from every old coordinate and give new-sector label j payoff $D_j + t$. Every full-team sum is unchanged. Equal sector sizes make the total change zero, and all coordinates are at least t, so this shifted lift also lies in the core.

Now let an old partial matching P have $r < m$ teams and worth $T - \delta$. Extend each team using new label equal to its v-label. This remains disjoint. If P uses every v-label with positive D_j, it collects all bonuses and keeps shortfall δ. The new unmatched set has $(k+1)(m-r)$ players. If the old floor witness has $\tau = \delta/[k(m-r)]$, the shifted witness attains the new floor $\delta/[(k+1)(m-r)]$. The unmatched budget proves that every unmatched coordinate is fixed to that value on the entire first optimal face.

An old unmatched target keeps ideal H if its diagonal team has worth H, its ideal is attained at an old core point, and the bonus at its diagonal label is zero. The tight new diagonal gives the upper bound; the unshifted lift of the attaining point gives the lower bound. More generally every old ideal has at least its previous value by unshifted lifting. No equality of entire cores or preservation of all ideals is needed.

6.2 Applying the lemma repeatedly

Start with the triangle's edge B from a to c. Its positive columns are $\mathcal{H} = \{O_s, L_s : 1 \leq s \leq q\}$ and their maxima are two. Set $D_j = 2$ on $\mathcal{H}$ and zero otherwise. The critical matching uses exactly these labels in sector c, so it collects every bonus. Also $D_{L_0} = 0$, preserving the target ideal one. Each subdivision adds $4q$ to the total worth and keeps the unit shortfall.

Subdivide the edge still carrying B, oriented from a to its current neighbor, repeatedly. The matching labels at that neighbor remain exactly $\mathcal{H}$, and all target diagonal labels remain L_0. Thus every transfer hypothesis propagates. After $\ell - 3$ steps the sector support is an ℓ-cycle, with

$$T_\ell = 6q + 1 + 4q(\ell - 3), \quad w(P) = T_\ell - 1, \quad |U| = \ell(q+1), \quad x_{c_{L_0}} = \frac{1}{\ell(q+1)}, \quad H_{c_{L_0}} = 1.$$

All pair entries remain in $\{0, 1, 2\}$ and the shifted floor witness is strictly positive.

Adding a zero-interaction sector leaves full-team worths unchanged. An old core point lifts by zero; a floor witness is shifted exactly as above with $D = 0$. Extend the critical matching by any r distinct new labels. The same unit budget and first-face pinning arguments apply; the target's diagonal worth and attainable ideal stay one.

Choose a simple cycle in any fixed cyclic allowed graph G. Embed the cycle construction there, add the other sectors with zero interactions, and put zero matrices on all other allowed edges,

including chords. This is allowed by the model and gives the stated leximin target $1/[K(q+1)]$ with ideal one.

For simultaneous protection retain the $2q+2$ original b, c players in U. Their old ideal-one witnesses lift without changing those payoffs; they remain unmatched. Thus their enlarged-core ideals are at least one, and (9) gives $\rho(M_{G,q}) \leq 1/(2q+2)$. We do not assert equality for longer cycles. Letting q grow gives $R(G) = 0$, while nonnegativity and $c(G) \leq R(G)$ give $c(G) = 0$. Together with the tree proof, this proves Theorem 2.1.

7 Evidence, scope, and contribution boundaries

The accompanying verification package preserves the earlier rational certificates and exact-solver checks for the component theorems. Tree checks compare the cut representation and clipping against raw full-team constraints. Cyclic checks verify integral matching and cover totals, unit shortfalls, positive floors and ideal attainment. The simultaneous checks reconstruct simplex vertices, all ideals, and the optimal-protection witnesses, and test both directions of the simplex description against raw cores for selected parameters. Negative controls reject damaged certificates. The package README states exact finite coverage and commands. These checks corroborate the all-parameter proofs above; they do not replace them or constitute a proof-assistant formalization.

The mathematical additions to the earlier chain–cycle manuscript are the branching-tree clipping proof, selective cyclic transfer, and the entire-core simplex and simultaneous-protection analysis. The older triangle family and classical tools are retained with their original provenance. Neither simultaneous ideal protection nor the bilateral half guarantee is proposed as a new fairness concept. The specific strict full-team graph classification is the proposed contribution, pending specialist priority assessment; original Quint, Lucas and Stuart comparisons remain incomplete under the free-only source constraint.

The classification does not cover productive partial-sector coalitions, unequal sector sizes, strictly positive matrices on every allowed edge, ratio-normalized leximin, or a selected proper core face. It does not determine the best finite-size simultaneous constant for all markets. The separate protected-subsidy program permits allocations above grand worth and is not a theorem of this paper. No journal submission, human referee endorsement, or confirmed field-level breakthrough is claimed.

Reproducibility and assistance. The manuscript and accompanying checks were developed with AI assistance. Scoped cross-model review and coordinator verification are documented in the package review record, separately from external human review. The latter and historical novelty assessment remain pending.

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